AIMing: Interactive Conjecture Proving

NSF 24-554: Artificial Intelligence, Formal Methods, and Mathematical Reasoning

Overview

The objective of this project is to integrate advanced artificial intelligence (AI) technologies with formal methods to enhance the functionality of interactive theorem provers (ITPs) in the mathematical reasoning process. By building upon recent advancements in Large Language Models (LLMs), we propose MathScy, a system that assists with conjecture formulation, proof construction, and counterexample finding. Current limitations of ITPs and formal methods include challenges in scalability, adaptability, and the depth of reasoning required to handle complex mathematical problems. MathScy is a compositional and modular system that uses fine-tuned Mixture-of-Expert LLMs for mathematical reasoning and integrates on-demand access to external mathematical tools. To ensure the developed tools are intuitive and accessible to mathematicians, the project will be guided by the principles of Human-Centered AI and Explainable AI. This will involve incorporating domain specific human feedback and intuition into the AI-assisted discovery process and generating human-understandable explanations of the AI's reasoning steps. To support this research, the project will create new standardized representations and datasets, as well as challenge problems tailored for benchmarking and advancing mathematical reasoning systems. The resulting techniques will be evaluated on significant open problems in mathematics and theoretical computer science. The project will focus on current research in the research areas of Applied, Combinatorial, and Computational Algebraic Geometry.

Intellectual Merit

The proposed research aims to enhance human-centered AI and interactive theorem proving (ITP) to automate mathematical discovery processes. It will develop compositional and modular models to integrate AI with ITPs for better mathematical reasoning. The project focuses on improving AI's mathematical reasoning capabilities through techniques in mathematical language processing and knowledge assimilation, enhancing conjecture discovery, proof search guidance, and counterexample identification. It will also design user-friendly tools for mathematicians by incorporating human feedback, creating new datasets and challenging problems for broader research use. Additionally, this research will concentrate on Applied, Combinatorial, and Computational Algebraic Geometry, leveraging AI and formal methods to address problems in these fields and their applications to STEM sciences.

Broader Impacts

MathScy aims to impact education, society, and science by providing accessible mathematical tools, enhancing education with new teaching methodologies, boosting industries reliant on complex modeling, and promoting inclusivity by collaborating with institutions serving underrepresented groups. The project will empower scientists in diverse areas of theoretical and applied mathematics, fostering collaboration to address problems in Applied, Combinatorial, and Computational Algebraic Geometry, and generating/proving/disproving conjectures in these fields.

Project Description

Introduction

Mathematical reasoning, encompassing conjecture formulation, proof construction, and counterexample finding, is critical to scientific and technological advancements. Traditionally, this reasoning process has been manual, requiring extensive human expertise. Recent advancements in artificial intelligence (AI) offer the potential to augment this process, yet current AI systems and interactive theorem provers (ITPs) face significant challenges that limit their efficacy in mathematical reasoning [1]. Per National Academies of Sciences, Engineering, and Medicine workshop on Artificial Intelligence to Assist Mathematical Reasoning, some of the identified challenges are the lack of data, scalability, depth of reasoning, adaptability, and explainability of current methods [2].

One major challenge in integrating AI systems into mathematical reasoning is scalability [3], as AI models often struggle to process and reason over large-scale mathematical structures, encompassing numerous definitions, theorems, and interconnected concepts. For example, reasoning about complex algebraic structures like groups, rings, and fields involves keeping track of many axiomatic properties and their logical implications. Another issue is the depth of reasoning [4]; AI models excel in pattern recognition but struggle with the logical reasoning needed for complex proofs and identifying subtle counterexamples. Furthermore, AI systems face adaptability challenges [5], as they are generally rigid and require significant manual intervention to adjust to new problems. For instance, an AI trained on algebra problems may not easily transfer to differential geometry without substantial modifications. Finally, explainability is crucial [6]; AI processes and outputs must be interpretable by human users. Current AI systems often function as "black boxes," hindering trust and acceptance among mathematicians, who need to understand the rationale behind AI-proposed conjectures or proofs to evaluate their validity and significance.

The proposed project, MathScy, aims to address these limitations by developing an agentic system that is compositional and modular. The system integrates an interactive multi-agent approach composed of three fine-tuned Mixture-of-Experts (MoE) [7] large language models (LLMs) with tool use capabilities and Retrieval Augmentated Generation (RAG) techniques [8]. Our approach considers fine-tuning MoE models that are specialized in math reasoning tasks [9]. In particular, we consider the use of models [9] that already surpass the performance of LEAN 4 translation and LEAN 4 theorem proving [10]. This integration is intended to enhance the scalability, depth of reasoning, adaptability, and explainability of AI systems in mathematical reasoning.

MathScy's architecture addresses scalability by using a Mixture-of-Expert approach, dynamically allocating computational resources to specialized sub-models for various mathematical domains, thus reducing computational load. This approach has demonstrated improvements in mathematical reasoning problems [9]. This approach also enhances adaptability by deploying models with differentiated expertise. To improve reasoning depth, MathScy combines LLMs for natural language processing and preliminary pattern recognition with formal reasoning engines for rigorous proof construction and counterexample identification. For instance, in topology, LLMs suggest potential properties and conjectures, which formal reasoning engines then verify. To enhance explainability, MathScy integrates mechanisms for generating human-understandable explanations, providing detailed logs and visualizations of the AI's reasoning steps. This transparency is crucial for validating AI outputs and building trust among mathematicians.

Guided by Human-Centered AI principles, mathematicians will play an integral role throughout the development process, incorporating their feedback to refine models. Mathematicians will test prototypes, provide input that informs the AI's reasoning pathways and enhance its performance. This collaborative approach ensures the AI system evolves consistently with human expertise and

judgment. Additionally, MathScy will develop standardized representations and datasets, and challenge problems to advance and benchmark mathematical reasoning systems, addressing current gaps and laying the foundation for future research. By combining AI and formal methods, MathScy aims to enhance interactive theorem provers and automate mathematical discovery processes.

Background

Applied, Combinatorial, and Computational Algebraic Geometry This project will focus on current research on Applied, Computational, and Enumerative Algebraic Geometry. The aim is to generate/prove/disprove conjectures in these fields. Algebraic geometry has a distinguished presence in the history of mathematics. Advances in computing over the last decades have revolutionized the area, making many problems tractable and providing a fertile ground for experimentation and conjecture. This proposal aims to push the envelope further using AI and formal methods to study problems arising in algebra, combinatorics, and symbolic computation and their applications in the sciences. Many important problems in the applied sciences (computer vision [11], statistics [12], spectral theory [13], systems biology [14, 15, 16], phylogenetics [17], chemical reaction networks [18]) can be expressed by polynomial equations in variables and parameters. Frequently, theoretical questions about these systems of polynomials and the algebraic and geometric objects they define (ideals and varieties) have significant implications in the motivating applications. Typical questions are: How many (real) solutions does the system have? What are the degree, dimension and other algebraic invariants (maximum likelihood degree, Euclidean distance degree) of the corresponding varieties, what is a minimal generating set of the defining ideal, or its Gröbner bases? These questions are usually approached using combinatorial, topological, geometric, and symbolic computation tools. Our aim is to leverage the power of AI and formal methods to study concrete problems in these fields.

Interactive Theorem Provers and Formal Methods Interactive theorem provers (ITPs) and formal methods have played a significant role in the advancement of mathematical reasoning and the verification of complex systems. ITPs, such as Coq, Isabelle, and HOL Light, provide a rigorous framework for defining mathematical concepts, formulating theorems, and constructing proofs [19]. These tools rely on formal language and a set of inference rules to ensure the correctness and consistency of mathematical statements. Formal methods, on the other hand, encompass a wide range of techniques and tools for the specification, development, and verification of software and hardware systems [20]. These methods employ mathematical logic and reasoning to establish the correctness and reliability of complex systems. Despite the success of ITPs and formal methods in various domains, they still face challenges in terms of scalability, automation, and usability [21]. The process of formalizing mathematical concepts and constructing proofs can be time-consuming and require significant expertise, limiting the widespread adoption of these tools. Moreover, the automation capabilities of ITPs are often limited, requiring manual intervention and guidance from the user. Researchers have been exploring ways to enhance the automation and usability of ITPs through the integration of techniques from artificial intelligence and machine learning [22].

AI and Large Language Models in Mathematical Reasoning Artificial intelligence (AI) techniques have shown promising results in enhancing the capabilities of mathematical reasoning systems. Machine learning approaches, such as neural networks and reinforcement learning, have been applied to various tasks in mathematical reasoning, including premise selection, proof search, and theorem proving [23]. These techniques aim to learn patterns and strategies from large datasets of mathematical proofs and use this knowledge to guide the reasoning process. In recent years, the emergence of large language models (LLMs) has revolutionized the field of natural language processing and opened new opportunities for mathematical reasoning [24]. LLMs, such as GPT-3 and BERT, are trained on vast amounts of text data and have demonstrated remarkable capabilities

in understanding and generating human-like text. Researchers have explored the application of LLMs to mathematical reasoning tasks, such as problem solving, theorem proving, and concept learning [25]. By fine-tuning LLMs on mathematical corpora and incorporating domain-specific knowledge, these models have shown promising results in generating conjectures, proposing proof strategies, and finding counterexamples. However, the integration of LLMs into formal reasoning systems poses several challenges. LLMs often lack the ability to capture the precise semantics and logical structure of mathematical statements, leading to inconsistencies and errors in generated proofs [26]. Moreover, the interpretability and explainability of LLMs remain open problems, making it difficult for mathematicians to understand and trust the reasoning process. Addressing these challenges requires the development of novel techniques for aligning LLMs with the formal reasoning framework and incorporating mechanisms for explanation and interaction.

Human-Centered AI and Explainable AI Human-Centered AI (HCAI) and Explainable AI (XAI) have emerged as crucial paradigms in the development of AI systems that are transparent, trustworthy, and aligned with human values and needs. HCAI focuses on designing AI systems that prioritize human interaction, collaboration, and empowerment by considering factors such as usability, interpretability, and ethical considerations throughout the AI development process, actively involving domain experts and end-users in the design and evaluation of AI systems to create tools that are intuitive, accessible, and responsive to human needs [27, 28]. XAI aims to develop AI systems that can provide meaningful explanations of their decision-making process and reasoning steps, which is crucial for building trust and facilitating effective collaboration between human experts and AI assistants in the context of mathematical reasoning, with approaches such as rule-based explanations, visual explanations, and natural language explanations being explored in various domains [29, 30]. Integrating HCAI and XAI principles into the development of AI-assisted mathematical reasoning systems presents challenges, such as striking a balance between automation and human control, and generating explanations that are both mathematically rigorous and understandable to users with varying levels of expertise, requiring the development of novel explanation techniques that can adapt to the user's background knowledge and provide insights at different levels of abstraction [31, 32]. Despite these challenges, the incorporation of HCAI and XAI principles in mathematical reasoning systems has the potential to revolutionize the way mathematicians interact with AI tools.

Methodology

MathScy System Architecture

The MathScy system is designed to be a comprehensive and modular solution for mathematical reasoning tasks, including conjecture formulation, proof construction, and counterexample finding. MathScy architecture consists of several key components that work together to provide an efficient, reliable, and explainable system for mathematicians. Figure 1 illustrates the high-level architecture of the MathScy system. At the core of MathScy are three specialized models, each fine-tuned using a Mixture-of-Experts approach based on Large Language Models (LLMs). These models are designed to handle specific tasks per each expert layer such as conjecture formulation, proof construction, and counterexample finding. Each model is equipped with tool use capabilities, allowing them to generate commands and interface with various external systems, services, and programs relevant to their respective tasks. To facilitate collaboration and decision-making among the specialized models, MathScy employs a multi-agent approach. The models communicate by exchanging messages and use multiple decision-making techniques to reach a consensus on the solution [33]. This multi-agent system ensures that the strengths of each model are leveraged and that the final solution is robust and reliable.

An explainable layer, powered by another LLM trained to provide explanations tailored to different audiences, is incorporated into the MathScy architecture. This layer is crucial for making the system's reasoning process transparent and understandable to mathematicians, promoting trust and adoption. Orchestrating the workflow of user requests is a key component of the MathScy system. An orchestration mechanism, potentially another LLM, is designed to handle requests from multiple mathematical domains. It determines which agents should be invoked, in what order, and manages the flow of information between them. The orchestration mechanism ensures that the appropriate external tools are used by each agent, monitors the debate and consensus process, validates intermediate results, and determines the final outcome.

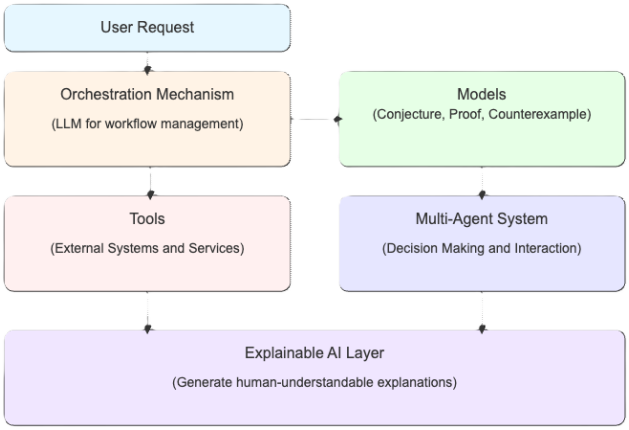

Figure 1: MathScy System Architecture

Compositional and Modular Design The MathScy system will be built upon a compositional and modular architecture to ensure flexibility, scalability, and maintainability. The core components of the system will include specialized models for conjecture formulation, proof construction, and counterexample finding, each built upon fine-tuned Mixture-of-Expert LLMs. These models will be designed to work independently on their respective tasks while also having the ability to collaborate and communicate with each other through a well-defined multi-agent framework. The modular design will allow for the seamless integration of new models, tools, and techniques as the system evolves, ensuring that MathScy remains at the forefront of AI-assisted mathematical reasoning research. The architecture will also incorporate an explainable AI layer, responsible for generating clear and intuitive explanations of the system's reasoning process and results. This layer will be designed to adapt the level of detail and complexity of explanations based on the user's background and expertise, ensuring that the system remains accessible and transparent to a wide range of users. Additionally, an orchestration mechanism will be implemented to manage the workflow of user requests and coordinate the invocation and interaction of the specialized models. This mechanism will be optimized for efficiency, effectiveness, and scalability, ensuring that MathScy can handle a large volume of complex mathematical reasoning tasks across various domains.

Integration of AI and Formal Methods A key challenge in developing MathScy is the effective integration of AI technologies, particularly LLMs, with formal methods. LLMs excel at natural language understanding and generation but often lack the rigorous logical reasoning capabilities required for mathematical proofs. On the other hand, formal methods provide a strong foundation for logical reasoning but can be difficult to apply to complex, real-world mathematical problems [34]. To bridge this gap, we will investigate several approaches for integrating AI and formal methods within the MathScy system. One approach will be to develop hybrid reasoning techniques that combine the strengths of LLMs and formal methods [35]. For example, LLMs can be used to generate high-level proof sketches or identify relevant axioms and theorems, which can then be rigorously verified and refined using formal methods. Another approach will be to incorporate formal methods directly into the training process of the LLMs, allowing them to learn logical reasoning patterns and constraints from formal proofs. This can be achieved through techniques such as knowledge distillation, where a formal reasoning system is used to generate a large dataset of proofs, which are then used to fine-tune the LLMs [36]. We will also explore the use of differentiable theorem provers, which are neural network-based systems that can be trained to perform logical

reasoning in a way that is compatible with gradient-based optimization methods used in deep learning.

Fine-Tuned Mixture-of-Expert LLMs for Mathematical Reasoning At the core of MathScy are the specialized models for conjecture formulation, proof construction, and counterexample finding, each built upon fine-tuned Mixture-of-Expert LLMs. Mixture-of-Expert models consist of multiple expert networks, each specializing in a particular sub-task or domain, and a gating network that learns to route input data to the most appropriate expert based on the task at hand [37]. This architecture allows for more efficient learning and improved performance compared to single-model approaches, particularly in complex and diverse domains such as mathematical reasoning. To develop these specialized models, we will first curate large-scale datasets of mathematical conjectures, proofs, and counterexamples from various sources, such as research papers, textbooks, and online repositories. These datasets will be carefully annotated and structured to facilitate the training and evaluation of the LLMs. We will then investigate various architectures and training strategies for the Mixture-of-Expert LLMs, such as the number and size of expert networks, the design of the gating network, and the use of task-specific loss functions and regularization techniques.

One key innovation in our approach will be the integration of tool use capabilities into the specialized LLMs [38]. Mathematical reasoning often involves the use of external tools and systems, such as computer algebra systems, theorem provers, and numerical solvers. By equipping the LLMs with the ability to generate commands for executing these tools and interpreting their outputs, we aim to create a more powerful and versatile mathematical reasoning system. To achieve this, we will explore techniques such as reinforcement learning, where the LLMs are trained to generate tool commands that maximize a reward function based on the quality and efficiency of the solutions.

Multi-Agent Approach for Collaborative Reasoning To enable effective collaboration and decision-making among the specialized models, we will develop a multi-agent approach based on advanced decision-making techniques and a flexible communication framework [39]. The multi-agent system will allow the models to exchange information, debate, and reach consensus on solutions to complex mathematical problems, leveraging their complementary strengths and expertise. We will investigate and apply various decision-making techniques, such as voting mechanisms, argumentation frameworks, and consensus algorithms, to facilitate the interaction among the specialized models. For example, the models could engage in a round of argumentation, where each model proposes a solution and provides supporting evidence, followed by a voting phase where the models collectively decide on the most promising solution based on the strength of the arguments. Alternatively, the models could use a consensus algorithm, such as Paxos or Raft, to converge on a solution through a series of proposals and feedback rounds. To support the multi-agent interaction, we will design and implement a flexible and scalable communication framework. This framework will define a common language and protocol for the models to exchange messages, share intermediate results, and coordinate their activities. The communication framework will be designed to handle a large volume of messages efficiently, ensure the consistency and integrity of the shared information, and handle failures or disagreements among the models.

Orchestration Mechanism To manage user requests and coordinate specialized models, we will design an orchestration mechanism as the central hub of the MathScy system. This mechanism will receive user queries, decompose them into mathematical reasoning tasks, allocate these tasks to the appropriate models, and compile the results into a comprehensive solution. It will use advanced techniques for parsing user requests in various formats, extracting key components like problem statements and desired outputs. The mechanism will map these components onto a standardized representation for the models. It will then determine the optimal sequence of model invocations, considering task dependencies, computational costs, and user constraints, using scheduling and planning algorithms to efficiently allocate tasks and adapt based on performance and feedback.

Evaluation and Validation To ensure MathScy's effectiveness, reliability, and usability, we will conduct rigorous evaluations using both quantitative and qualitative measures. We will assess the system's performance, explanation quality, and user satisfaction through a comprehensive suite of benchmark problems and datasets from various mathematical domains. These benchmarks will test the accuracy, efficiency, and robustness of the specialized models and orchestration mechanism, comparing MathScy's performance against state-of-the-art baselines and human experts. Additionally, we will conduct user studies with mathematicians and students to gather qualitative feedback through evaluations, questionnaires, interviews, and focus groups, refining the system iteratively.

Human-Centered AI And Explainable AI Techniques

Human-in-the-loop Approaches Incorporating human feedback and intuition is crucial for developing an AI-assisted mathematical reasoning system that is truly human-centered and aligned with the needs and expectations of mathematicians. We will actively engage with the mathematical research community throughout the development process, seeking their input and feedback to guide the design and evaluation of the MathScy system. We will employ participatory design [40], which involves involving mathematicians as co-designers and collaborators in the development process, conducting workshops, interviews, and focus groups to understand their workflows, challenges, and aspirations for AI-assisted mathematical reasoning, enabling us to identify key pain points and opportunities for innovation. We will also explore interactive machine learning, applying Human-in-the-loop approaches to actively learn from human feedback by developing user interfaces and interaction paradigms that allow mathematicians to provide feedback on the system's performance, such as indicating the relevance and quality of generated conjectures, proofs, and explanations, which will be used to fine-tune the AI models and adapt them to the specific domain knowledge and intuition of the mathematicians.

Explainable AI Approaches To ensure the transparency and interpretability of MathScy, we will develop an explainable AI layer that generates clear and intuitive explanations of the system's reasoning process and results. This layer will cater to users with diverse backgrounds and levels of expertise, from novice mathematicians to experienced researchers. We will investigate and implement various explanation generation techniques, such as natural language explanations, visual explanations (e.g., proof trees, graphs), and interactive explanations (e.g., step-by-step walkthroughs, and QA interfaces). These techniques will be based on state-of-the-art approaches in explainable AI, such as attention mechanisms, counterfactual reasoning, and example-based explanations. The choice of explanation technique will depend on the nature of the mathematical reasoning task, the complexity of the solution, and the user's preferences and goals. To generate high-quality explanations, the explainable AI layer will need to have a deep understanding of the reasoning process employed by the specialized models. To achieve this, we will explore techniques for aligning the explanations with the internal reasoning steps of the models, such as extracting intermediate proof states, identifying key decision points, and tracking the flow of information between the models. We will also investigate methods for quantifying and communicating the uncertainty associated with the generated explanations, helping users to calibrate their trust in the system's outputs. Another key aspect of the explainable AI layer will be its ability to adapt the level of detail and complexity of explanations based on the user's background and expertise. To achieve this, we will develop user modeling techniques that can infer the user's knowledge and skills from their interactions with the system (e.g., the types of queries they make, the feedback they provide, their speed and accuracy in comprehending the explanations). Based on this user model, the explainable AI layer will dynamically adjust the depth, sophistication, and presentation format of the explanations, ensuring that they are always relevant and accessible to the individual user.

Implementation

The implementation of the MathScy project involves a plan that integrates various technologies and methodologies to create the proposed system. The implementation process is structured to ensure robust development, testing, and deployment phases, allowing for iterative improvements and adaptability. The implementation plan is divided into five main phases: (1) System Design and Architecture, (2) Development of Core Components, (3) Interface Development and HITL Integration, (4) Testing, Validation, and Refinement, and (5) Deployment and Continuous Improvement. Each phase consists of several tasks that are designed to achieve specific objectives and outcomes, with well-defined metrics to measure progress and success.

Phase 1: System Design And Architecture

The first phase of the MathScy implementation focuses on establishing the foundation for the system by designing a modular and scalable architecture that can support the integration of various AI technologies and mathematical tools. This phase involves three main tasks: Architecture Development, Technology Stack Selection, and Dataset Development.

Task 1.1 Architecture Development

The Architecture Development task aims to create a modular and scalable system architecture. Objectives: (i) design a modular architecture for integrating LLMs, Mixture-of-Experts systems, and mathematical tools; (ii) ensure scalability and maintainability; (iii) establish clear interfaces between components. Activities: 1.1.1 analyze system requirements and constraints, 1.1.2 design high-level architecture defining main components and interactions, 1.1.3 create detailed component specifications, 1.1.4 validate architecture through peer reviews and stakeholder feedback, 1.1.5 document architecture using standard notations. Outcomes: (i) comprehensive system architecture document; (ii) well-defined interfaces and protocols; (iii) implementation roadmap. Metrics: (i) completeness and clarity of documentation; (ii) ability to support integration of AI and mathematical tools; (iii) stakeholder and reviewer feedback and approval.

Task 1.2 Technology Stack Selection

The Technology Stack Selection task focuses on identifying the most appropriate programming languages, frameworks, AI platforms, and computational resources for MathScy. Objectives: (i) evaluate and compare technology options based on performance, scalability, and compatibility; (ii) select technologies that support complex mathematical reasoning; (iii) ensure technologies are well-documented with strong community support. Activities: 1.2.1 survey state-of-the-art technologies in AI, formal methods, and mathematical reasoning, 1.2.2 define evaluation criteria and benchmarks, 1.2.3 set up test environments and conduct performance tests, 1.2.4 analyze results and decide on technology stack, 1.2.5 document selected technologies and configurations. Outcomes: (i) well-defined and optimized technology stack; (ii) best practices and guidelines for using selected technologies; (iii) plan for acquiring and setting up computational resources. Metrics: (i) performance and scalability in handling complex mathematical reasoning tasks; (ii) ease of integration and interoperability; (iii) level of community support and documentation.

Task 1.3 Dataset Development

The Dataset Development task involves creating diverse and representative datasets for training and evaluating the AI models. Objectives: (i) curate and annotate high-quality datasets from various mathematical sources; (ii) ensure diversity and inclusivity across mathematical domains, problem types, and difficulty levels; (iii) establish rigorous annotation guidelines and quality control mechanisms. Activities: 1.3.1 collaborate with mathematicians and computer scientists to identify and collect relevant sources, 1.3.2 develop custom ontologies and annotation schemas, 1.3.3 use NLP techniques to extract and structure mathematical knowledge, 1.3.4 set up annotation tools and platforms, 1.3.5 establish quality control processes and metrics, 1.3.6 document dataset

creation process and publish datasets under open licenses. Outcomes: (i) high-quality, diverse, well-annotated datasets for training and evaluating AI models; (ii) standardized annotation guidelines and quality control mechanisms; (iii) platform for collaborative dataset development. Metrics: (i) coverage and diversity across mathematical domains, problem types, and difficulty levels; (ii) accuracy and consistency of annotations; (iii) level of community engagement and contributions.

DEI 1: We are committed to promote diversity, equity, and inclusion in all aspects of its implementation. In this phase, we will actively seek input and feedback from a diverse group of stakeholders, including researchers, educators, and practitioners from underrepresented groups in mathematics and computer science. In the Dataset Development task, we will make conscious efforts to include mathematical knowledge from a wide range of cultures, languages, and historical periods, and to ensure that the datasets are free from biases and stereotypes.

Phase 2: Development Of Core Components

The second phase of the MathScy implementation focuses on the development of the core AI models and their integration with formal methods and interactive theorem provers. This phase involves three main tasks: AI Model Training, Integration of Formal Methods, and Evaluation and Refinement.

Task 2.1 AI Model Training

The AI Model Training task involves developing and fine-tuning LLMs and Mixture-of-Experts systems for mathematical reasoning. Objectives: (i) develop state-of-the-art AI models for accurate and efficient mathematical reasoning tasks; (ii) fine-tune models using Task 1.3 datasets and other relevant benchmarks; (iii) evaluate model performance using well-defined metrics and baselines. Activities: 2.1.1 design and implement LLM and Mixture-of-Experts architectures, 2.1.2 collect and preprocess additional datasets for fine-tuning, 2.1.3 set up distributed training pipelines and hyperparameter optimization frameworks, 2.1.4 conduct experiments to evaluate model performance, 2.1.5 analyze results and iteratively refine models. Outcomes: (i) highly accurate and efficient AI models for mathematical reasoning; (ii) comprehensive evaluation framework with metrics, baselines, and benchmarks; (iii) best practices and guidelines for fine-tuning and deploying models. Metrics: (i) accuracy and efficiency on mathematical reasoning tasks; (ii) generalizability and robustness on out-of-distribution and adversarial examples; (iii) scalability and maintainability of training pipelines and infrastructure.

Task 2.2 Integration of Formal Methods

The Integration of Formal Methods task focuses on integrating the trained AI models with formal methods and interactive theorem provers (ITPs). Objectives: (i) develop robust interfaces and protocols for integrating AI models with existing formal methods and ITPs; (ii) ensure correctness and consistency through extensive testing and verification; (iii) optimize performance and usability based on user feedback. Activities: 2.2.1 analyze input/output formats and APIs of target formal methods and ITPs, 2.2.2 design and implement adapters and wrappers for format conversion, 2.2.3 develop testing and verification frameworks for ensuring correctness and consistency. Outcomes: (i) well-defined and tested interfaces and protocols for AI and formal methods integration; (ii) comprehensive testing and verification framework for the integrated system. Metrics: (i) accuracy and efficiency in solving mathematical reasoning tasks; (ii) user satisfaction and adoption level.

DEI 2: In the Development of Core Components phase, we will strive to create models that are free from biases and stereotypes, and that can handle mathematical reasoning tasks from a wide range of domains and cultures. We will also ensure that the interfaces and protocols for integrating the AI models with formal methods are accessible and usable by researchers and practitioners from diverse backgrounds and skill levels. In the evaluation and refinement of the models and integrated system, we will actively seek feedback and input from a diverse group of mathematicians and users,

including those from underrepresented groups in mathematics and computer science.

Phase 3: Interface Development And HITL Integration

The third phase of the MathScy implementation focuses on the development of user-friendly interfaces and the integration of human-in-the-loop (HITL) systems for collaborative mathematical reasoning. This phase involves three main tasks: User Interface Design and Explainable AI, Human-in-the-Loop Systems, and Expert Evaluation Test Phase.

Task 3.1 User Interface Design and Explainable AI

The User Interface Design and Explainable AI task involves developing intuitive user interfaces and integrating explainable AI techniques for transparent reasoning processes. Objectives: (i) design user interfaces enabling easy interaction for mathematicians at all expertise levels; (ii) integrate explainable AI for clear explanations of reasoning steps and outputs; (iii) iteratively improve usability and explainability through user studies and feedback. Activities: 3.1.1 conduct user research and gather requirements, 3.1.2 design wireframes, mockups, and prototypes, 3.1.3 implement user interfaces using modern web technologies, 3.1.4 integrate explainable AI techniques, 3.1.5 conduct user studies and usability tests. Outcomes: (i) intuitive and accessible user interfaces; (ii) explainable AI techniques providing clear explanations tailored to user expertise. Metrics: (i) user satisfaction and adoption level; (ii) accuracy and understandability of explanations; (iii) efficiency and scalability in handling large-scale reasoning tasks.

Task 3.2 Human-in-the-Loop Systems

The Human-in-the-Loop Systems task focuses on developing HITL systems for mathematicians to provide real-time feedback and guidance to the AI models. Objectives: (i) design HITL systems allowing mathematicians to interactively guide and refine AI reasoning processes; (ii) develop efficient algorithms for incorporating human feedback and updating models in real-time; (iii) evaluate HITL effectiveness and usability through user studies and case studies. Activities: 3.2.1 analyze HITL requirements and use cases, 3.2.2 design and implement HITL architectures and protocols, 3.2.3 integrate HITL with user interfaces and explainable AI techniques, 3.2.4 conduct testing and validation using benchmarks and case studies, 3.2.5 iteratively refine HITL based on feedback and insights. Outcomes: (i) robust and efficient HITL systems for collaborative mathematical reasoning; (ii) evaluation framework based on user studies and case studies; (iii) best practices and guidelines for deploying HITL in real-world scenarios. Metrics: (i) accuracy and efficiency of HITL-assisted reasoning compared to AI alone; (ii) user satisfaction and engagement; (iii) scalability and robustness in handling diverse reasoning tasks.

Task 3.3 Expert Evaluation Test Phase

The Expert Evaluation Test Phase task involves recruiting diverse mathematicians and domain experts to test and evaluate MathScy, providing feedback for improvement. Objectives: (i) assemble a diverse group of mathematicians and domain experts; (ii) conduct rigorous evaluation using well-defined criteria and methodologies; (iii) gather feedback and incorporate into iterative refinement. Activities: 3.3.1 develop recruitment strategy for diverse experts, 3.3.2 design and implement evaluation methodology, 3.3.3 conduct evaluation through workshops, interviews, and surveys, 3.3.4 analyze feedback and prioritize areas for improvement, 3.3.5 communicate results to development team and incorporate feedback. Outcomes: (i) diverse group of actively engaged experts; (ii) comprehensive and rigorous evaluation; (iii) prioritized recommendations and areas for improvement. Metrics: (i) diversity and expertise of recruited experts; (ii) quality and thoroughness of evaluation; (iii) impact and feasibility of recommendations.

DEI 3: In the Interface Development and HITL Integration phase, we will ensure that the user interfaces are accessible and usable by mathematicians and users from diverse backgrounds and skill levels, including those with disabilities and non-native English speakers. We will also strive to

provide explanations and feedback that are culturally sensitive and avoid biases and stereotypes. In the recruitment and engagement of mathematicians and domain experts for the evaluation phase, we will actively seek participation from underrepresented groups in mathematics and computer science, including women, minorities, and researchers from developing countries.

Phase 4: Testing, Validation, And Refinement

The fourth phase of the MathScy implementation focuses on the comprehensive testing, validation, and refinement of the integrated system, to ensure its accuracy, efficiency, and robustness in solving real-world mathematical reasoning tasks. This phase involves two main tasks: System Performance Testing, and Validation with Real-world Problems.

Task 4.1 System Performance Testing

The System Performance Testing task involves designing and executing a comprehensive testing plan to assess MathScy's performance, scalability, and reliability. Objectives: (i) develop a testing plan covering all key components and functionalities; (ii) execute the plan using well-defined test cases, datasets, and benchmarks, collecting performance metrics and logs. Activities: 4.1.1 define testing scope and objectives, 4.1.2 design and implement test cases and datasets, 4.1.3 set up testing infrastructure and environments, 4.1.4 execute testing plan and collect performance metrics and logs, 4.1.5 analyze results and identify performance bottlenecks, errors, or inconsistencies. Outcomes: (i) comprehensive and well-documented testing plan; (ii) detailed performance metrics and logs providing insights into accuracy, efficiency, and robustness. Metrics: (i) coverage and completeness of testing plan; (ii) accuracy and reliability of performance metrics and logs; (iii) relevance and impact of identified areas for improvement and optimization.

Task 4.2 Validation with Real-world Problems

The Validation with Real-world Problems task involves applying MathScy to real-world mathematical reasoning problems and datasets to assess its effectiveness and practicality. Objectives: (i) identify and select diverse real-world mathematical reasoning problems and datasets; (ii) validate MathScy's effectiveness and practicality compared to existing methods and tools, and gather expert feedback. Activities: 4.2.1 conduct literature review and community outreach to identify representative problems and datasets, 4.2.2 set up and configure MathScy for selected problems and datasets, 4.2.3 run MathScy and collect performance and usability metrics, 4.2.4 compare MathScy's performance and usability with existing methods and tools. Outcomes: (i) diverse set of real-world mathematical reasoning problems and datasets; (ii) validated and refined MathScy system based on comparisons and expert feedback. Metrics: (i) diversity and relevance of selected problems and datasets; (ii) MathScy's performance and usability on problems and datasets; (iii) impact and significance of validation results in publications and community adoption.

DEI 4: In the Testing, Validation, and Refinement phase, we will strive to include problems and datasets from a wide range of mathematical domains and cultural contexts, and to ensure that they are accessible and relevant to a diverse community of mathematicians and users. We will also actively seek feedback and input from underrepresented groups in mathematics and computer science, to ensure that the MathScy system is meeting their needs and expectations. In the dissemination of the validation results, we will make efforts to reach out to a broad and diverse audience, including researchers, educators, and students from different countries and backgrounds.

Phase 5: Deployment And Continuous Improvement

The final phase of the MathScy implementation focuses on the deployment of the validated and refined system to a wider community of mathematicians and users, and the establishment of a continuous improvement process to ensure its long-term sustainability and impact. This phase involves three main tasks: Deployment, Feedback Loops and Updates, and Community Engagement.

Task 5.1 Deployment

The Deployment task involves preparing and releasing MathScy for public use, including infrastructure setup, documentation, and user support. Objectives: (i) prepare MathScy for public release, including software packaging, hosting infrastructure setup, and user documentation; (ii) deploy MathScy to a secure and scalable hosting environment, ensuring availability and reliability. Activities: 5.1.1 package MathScy into distributable software components, 5.1.2 set up hosting infrastructure, including servers, storage, networking, security, monitoring, and backup, 5.1.3 create user documentation and tutorials, 5.1.4 deploy MathScy to hosting infrastructure and thoroughly test for availability, reliability, and performance. Outcomes: (i) packaged and distributable MathScy system ready for installation and use; (ii) secure, scalable, and reliable hosting infrastructure supporting large numbers of users and requests. Metrics: (i) ease and reliability of installation and configuration; (ii) availability, performance, and security of hosting infrastructure; (iii) user satisfaction and engagement.

Task 5.2 Feedback Loops and Updates

The Feedback Loops and Updates task involves establishing a continuous feedback and improvement process for MathScy based on user input and insights. Objectives: (i) collect and analyze user feedback, bug reports, and feature requests, prioritizing based on impact and feasibility; (ii) design and implement a roadmap for continuous improvement and evolution based on prioritized feedback and project goals. Activities: 5.2.1 set up feedback collection and management system, encouraging user input, 5.2.2 regularly review and analyze feedback, categorizing and prioritizing issues and requests, 5.2.3 design and implement improvement roadmap outlining planned updates, features, and optimizations, 5.2.4 develop and test updates and patches, ensuring compatibility and stability. Outcomes: (i) comprehensive and up-to-date collection of user feedback, providing insights into strengths, limitations, and potential improvements; (ii) well-defined and actionable roadmap for continuous improvement based on prioritized feedback and project goals. Metrics: (i) quantity, quality, and diversity of collected feedback; (ii) coherence, feasibility, and impact of improvement roadmap; (iii) adoption, satisfaction, and performance of released updates and patches.

Task 5.3 Community Engagement

The Community Engagement task involves active outreach, communication, and collaboration with the broader mathematical reasoning community to promote MathScy's adoption, contribution, and sustainability. Objectives: (i) raise awareness and understanding of MathScy among the mathematical reasoning community from diverse domains and backgrounds; (ii) foster ownership and contribution among users and stakeholders, encouraging active participation in development, testing, and promotion. Activities: 5.3.1 develop and execute communication and dissemination plan, including project website, social media, and regular publications, 5.3.2 organize and participate in community events and activities, inviting users to contribute and showcase work, 5.3.3 establish and moderate online and offline forums and channels for users to connect, collaborate, and support each other, 5.3.5 develop and provide resources and incentives for users to contribute to development, testing, and promotion. Outcomes: (i) large, diverse, and engaged community of users actively participating in development, testing, and promotion, providing feedback, contributions, and support; (ii) strong brand and reputation for MathScy as a leading and innovative platform for mathematical reasoning, attracting new users, partners, and funding opportunities. Metrics: (i) size, diversity, and engagement of user community; (ii) visibility, recognition, and influence of MathScy; (iii) number, scope, and impact of partnerships and collaborations.

DEI 5: In this phase, we will strive to reach out to and collaborate with a wide range of user groups and communities, including those from underrepresented and marginalized backgrounds, and to provide them with accessible, relevant, and empowering resources and opportunities. We will also ensure that the communication, documentation, and user interface of the MathScy system are inclusive, respectful, and culturally sensitive, and that they do not perpetuate or reinforce any

biases, stereotypes, or discriminatory practices. In the organization and participation of community events and activities, we will make conscious efforts to invite and feature speakers, presenters, and participants from diverse backgrounds and perspectives, and to create welcoming, equitable, and inclusive environments for learning, networking, and collaboration.

Milestones And Success Criteria

Phase	Task Objectives	Methods/ Metrics	Outcome Measures
Phase 1: System Design and Architecture	1.1 Design a modular architecture that supports integration of LLMs, external mathematical tools and databases	Feedback and approval from project stakeholders and external reviewers	A comprehensive system architecture document and a roadmap for implementation of the system
	1.2 Evaluate and compare different technology options based on performance and scalability	Performance and scalability of selected technologies	A well-defined and optimized technology to support development of MathScy system
	1.3 Curate and annotate high quality datasets of mathematical knowledge	Coverage and diversity of created datasets	High-quality, diverse and well-annotated datasets
Phase 2: Development of Core Components	2.1 Develop state-of-the-art AI models with the ability to perform mathematical reasoning	Accuracy and efficiency of trained models on mathematical tasks	Efficient AI models with the guidelines for fine-tuning and deploying models in real world settings
	2.2 Develop robust interfaces and protocols for integrating AI models with formal methods	Accuracy and efficiency of integrated system in solving mathematical reasoning task	Set of well-defined and thoroughly tested interfaces and protocols for integrating AI models with formal methods and ITPs
Phase 3: Interface Development & HITL Integration	3.1 Design user interfaces that are easy for mathematicians to use	User satisfaction survey	Intuitive and accessible user interface
	3.2 Design HITL systems that allow mathematicians to interactivity guide and refine AI reasoning	User engagement and accuracy survey	Robust and efficient HITL systems for collaborative mathematical reasoning
	3.3 Expert evaluation test phase	Survey and feedback from domain experts, impact and feasibility of recommendations	Prioritized recommendations and areas for improvement
Phase 4: Testing, Validation & Refinement	4.1 Develop a testing plan covering all key components and functionalities	Accuracy and reliability of performance metrics and logs	Comprehensive and well-documented testing plan
	4.2 Identify and select diverse real-world problems to test effectiveness of MathScy	MathScy performance and usability on problems and datasets	A diverse set of real-world mathematical reasoning problems and datasets
Phase 5: Deployment and Continuous Improvement	5.1 Prepare MathScy for public release	User satisfaction and engagement survey	Packaged and distributable MathScy system ready for installation and use
	5.2 Collect and analyze user feedback, bug reports, feature requests to improve MathScy	Quality, quantity, and diversity of feedback and performance of updates	Well-defined and actionable roadmap for continuous improvement based on feedback
	5.3 Raise awareness and understanding of MathScy among mathematicians	Visibility, recognition and influence of MathScy through number of partnerships	Large, diverse, and engaged community of users actively participating in development, testing, and promotion of MathScy

Figure 2: Methods, Measures and Metrics of research phase performance

The abridged evaluation rubric in Figure 2 provides the high-level objectives of each task, the methods and metrics that will be used to evaluate the research, and the outcome measures that will be reported to the research teams in the short term (formative feedback) and the long term (summative feedback). Validated surveys will provide qualitative feedback to be analyzed with quantitative experimental measurement data (performance metrics for the LLM model).

Project Management And Collaboration Plan

The project will be managed through a layered approach whereby DEI, External advisory board, and research functions interface with each other and with an overall project management function as shown in Figure 3, with responsibilities and backup.

Final project oversight will be managed by PI Hubig, with co-PI Toxtli handling overall program activities, budget, and reporting. Weekly hybrid meetings will ensure team coordination, using collaboration tools like MS Teams for communication and information sharing. Domain experts will define the feasibility and usability of MathScy architecture, providing feedback to fine-tune the model. Each expert will be directly involved for approximately 10 days and indirectly as needed for tool testing. Their input will influence research phases, ensuring the architecture is widely adopted by mathematicians. Further details on domain experts are in the Facilities, Equipment, & Other Resources document. The overall timeline for the project is provided in Figure 4.

Diversity, Equity, and Inclusion. Activities related to access, inclusion, and equity within the program itself, through the research carried out will be governed by the DEI team.

Convergent Research. Each research phase will be the responsibility of a Phase Leader working

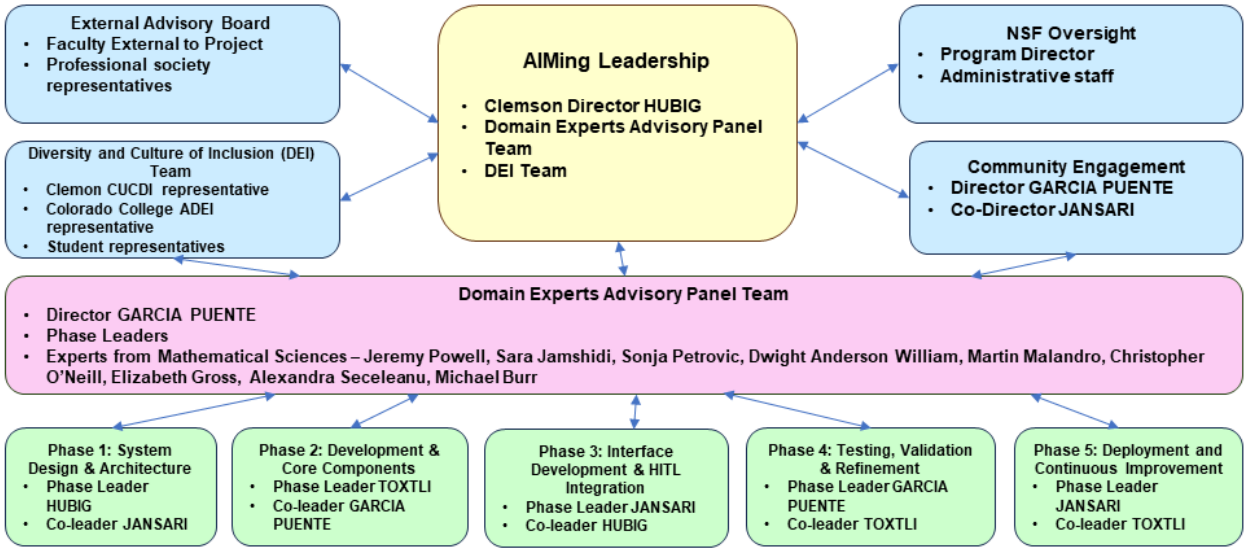

Figure 3: MathScy Organization: Each research phase has a lead and backup; composed of directors, committee leaders interfacing with NSF, External Advisory Board, and Panel of Experts

together with a co-leader who will support and assume the chair's place in case of attrition from the program. Researcher Hubig will lead Research Phase 1 (System Design and Architecture); researcher Toxtli will lead Phase 2 (Development & Core components); researcher Jansari will lead Phase 3 (Interface Development & HITL Integration); researcher Garcia Puente will lead Phase 4 (Testing, Validation & Refinement), and researcher Jansari will lead Phase 5 (Deployment and Continuous Improvement). The leaders of each phase will meet quarterly to share and integrate detailed findings.

Intellectual Merit

The MathScy project leverages cutting-edge AI technologies like large language models, Mixture-of-Experts methods, AI-driven tool use, and multi-agent systems to advance artificial intelligence, formal methods, and mathematical reasoning. By automating tasks such as conjecture formulation, proof construction, and counterexample finding, MathScy aims to revolutionize mathematical research and enable new discoveries. Its modular system design fosters collaboration across AI, computer science, and mathematics, promoting the development of innovative tools and techniques. Emphasizing human-centered and explainable AI, MathScy ensures accessibility for mathematicians and students, transforming educational methodologies. The project also sets new standards for AI-assisted mathematical reasoning tools, promoting trust and adoption within the research community, and creating standardized datasets and benchmarks for consistent evaluation in mathematical AI.

Broader Impacts

The MathScy project integrates AI technologies with formal methods to enhance interactive theorem provers, impacting beyond mathematical sciences. By aiding mathematicians in areas like algebraic geometry, combinatorics, and optimization, MathScy fosters collaboration and advances research. Its AI-powered tools streamline research, making complex concepts accessible to a broader audience and improving educational resources. The project's technologies can also benefit fields like data science, finance, and telecommunications by enhancing mathematical modeling and analysis. Additionally, MathScy aids policy-making by validating models used in economic forecasting

Phase	Task	Title	Responsible	24 Q4	25 Q1	25 Q2	25 Q3	25 Q4	26 Q1	26 Q2	26 Q3	26 Q4	27 Q1	27 Q2	27 Q3
Phase 1: System Design & Architecture	Task 1.1	Architecture Development	Hubig/ Toxtli												
	Task 1.2	Technology Stack Selection	Hubig/ Jansari												
	Task 1.3	Dataset Development	Garcia/ TBD												
Phase 2: Development of Core Components	Task 2.1	AI Model Training	Toxtli/ Hubig/ Jansari												
	Task 2.2	Integration of Formal Methods	Toxtli/ Garcia												
Phase 3: Interface Development & HITL Integration	Task 3.1	User Interface Design & Explainable AI	Hubig/ Jansari												
	Task 3.2	Human-in-the-loop System	Toxtli/ Jansari												
	Task 3.3	Expert Evaluation Test Phase	Garcia/ TBD												
Phase 4: Testing, Validation & Refinement	Task 4.1	System Performance Testing	Hubig/ Toxtli												
	Task 4.2	Validation with Real-world Problems	Garcia/ Toxtli												
Phase 5: Deployment and Continuous Improvement	Task 5.1	Deployment	Toxtli/ Jansari												
	Task 5.2	Feedback Loops & Updates	Hubig/ Jansari												
	Task 5.3	Community Engagement	Garcia/ Jansari												

Figure 4: MathScy Timeline. Defined tasks, responsibilities, and checkpoints

and resource management. Emphasizing human-centered and explainable AI, it promotes systems that augment human intelligence and foster global collaboration through standardized datasets and benchmarks.

Results From Prior NSF Support

(1) PI Hubig currently serves as a Co-PI on the project "NSF EPSCOR: ADAPT: AI-enabled Biomedical Devices" NSF 1829008, ($20,993,421, 9/2024–8/2028). *Intellectual Merit*: New XAI-enabled biomedical devices in nine different areas from 4 institutions. *Broader Impacts*: This project is designed to encourage the development and implementation of novel biomedical devices embedded with XAI. (2) Co-PI Toxtli currently serves as a co-PI on the project "Collaborative Research: The Future of Aviation Inspection: Artificial Intelligence and Mixed Reality as Agents of Transformation" FoW-2326185, ($1,558,433, 9/11/23 - 9/11/27). *Intellectual Merit*: This project conducts research on: 1) Human-AI interaction; and 2) real-time maintenance aided by multimodal models. *Broader Impacts*: It will impact safety by improving the accuracy of periodic inspections. Publications: [41]. (3) Co-PI Garcia Puente currently serves as PI on the project "Conference:Workshop on Computational and Applied Enumerative Geometry" NSF-DMS 2419363 ($22,500, 05/2024-04/2025). *Intellectual Merit*: Enumerative geometry problems and their applications are fundamental to STEM fields including mathematics, particle physics, robotics, and computer vision.*Broader Impacts*: The NSF funding was targeted at supporting early career scientists and members of underrepresented groups in science, and will bring generate new collaborations to solve problems in the field of enumerative geometry.

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Facilities, Equipments, & Other Resources

Domain Experts Advisory Committee

For evaluation and validation purposes, we have assembled a group of researchers who work in a large spectrum of fields related to applied, combinatorial, and computational algebraic geometry. This group of nine mathematicians is formed by Michael Burr (Associate Professor of Mathematics at Clemson University); Elizabeth Gross (Associate Professor of Mathematics, University of Hawai'i at Manoa); Sara Jamshidi (Assistant Professor of Computer Science and Mathematics and Director of the Applied Data Center at Lake Forest College); Martin Malandro (Associate Professor of Mathematics at Sam Houston State University); Christopher O'Neill (Associate Professor of Mathematics at San Diego State University); Sonja Petrovic (Professor of Applied Mathematics at Illinois Institute of Technology); Jeremy Powell (AP Calculus AB/BC at TL Hanna High School); Alexandra Seceleanu (Associate Professor of Mathematics at the University of Nebraska-Lincoln); and Dwight Anderson Williams (Assistant Professor of Mathematics at Morgan State University). This group of four women and five men includes faculty at Hispanic Serving Institutions (Sam Houston State), Historically Black research universities (Morgan State), Pacific Island universities (University of Hawai'i at Manoa) and a collection of large state universities and liberal arts colleges.

Clemson University High Performance Computing Palmetto Cluster

https://www.palmetto.clemson.edu/palmetto/userguide_palmetto_overview.html

PI Hubig and Co-PI Toxtli have access to Clemson's high performance computing cluster, known as Palmetto, which was developed to serve the university's wide-ranging computational research needs. Developed by Clemson Computing and Information Technology (CCIT) in collaboration with faculty researchers across the university, Palmetto provides a shared platform that optimizes resources for all users. The HP-based Palmetto Cluster is designed to suit many different research applications, with many powerful multi-core nodes, each with a significant amount of memory. Palmetto currently ranks 244 in the world (TOP500 list, November 2017) and 5 among U.S. academic facilities that are not national centers. It can do 551 trillion calculations per second and currently comprises 23,452 cores and a Xeon E5-2665 8C 2.4GHz processor. The system also has multicore large-shared-memory systems, with 45,312 GB of memory and standard multicore distributed-memory configurations available. It has an operating system on all nodes: Scientific Linux 6 (based on RedHat Enterprise Linux) a PBS Professional 12.0 Job queuing system and is connected to Internet2's 100 GbE (gigabit Ethernet) Advanced Layer 2 Service. Each user account has 100 GB of backed-up space in the /home directory, and each user can store up to 1,000,000 files there. All users share a 218 TB temporary "work" space in /newscratch. The process is a Xeon E5-2665 8C 2.4GHz on an Infiniband FDR Interconnect. This system also contains 360 TB of large-scale, high performance disk storage and is housed at Clemson's Information Technology Center in a 24/7 monitored environment with proper power, cooling and physical security. Faculty/staff can purchase Palmetto nodes through what's commonly called a "condo" model where resource owners are buying preemption units accessed using a dedicated owner job queue that is associated with the resources purchased. Preemption units give an owner's job the ability to (1) preempt general work jobs to acquire requested compute resources, (2) prevent the owner's job from being preempted by other owner's jobs, and (3) run for up to 168 hours. An owner of Palmetto storage resources will be granted a dedicated user group that is used to govern access to the purchased storage space. An owner can request that any Clemson students, faculty, or staff be added to that owner's user group for access to the purchased storage space. An owner can also set up Visitor ID accounts for external collaborators that will enable those external collaborators to log-in to Palmetto and access the owner's purchased storage space. The Palmetto Cluster is made possible through a partner-

ship between faculty and information technology administrators. A shared funding formula makes Palmetto available to the entire Clemson community on a first-come, first-served basis; faculty researchers who require predictable computational availability may purchase nodes with reserved priority. The university provides additional nodes, hardware infrastructure, and system administration support. This gives individual researchers high performance computing capacity previously unavailable at Clemson and allows the university to compete for the research grants and contracts that are vital to Clemson's goals as a major research university.

Clemson Council For Diversity And Inclusion (CUCDI)

https://www.clemson.edu/inclusion/about/index.html

The primary goals of Clemson Inclusion Equity are to:

• Guide the University's diversity and inclusion agenda. Promote an inclusive and supportive campus and community environment.

• Identify and allocate appropriate resources to support University diversity initiatives.

• Promote the institution's commitment and accomplishments regarding diversity and inclusion to internal and external communities.

• Create innovative programs and projects that advance and support campus diversity and inclusion.

• Develop, implement and coordinate a comprehensive strategic plan to advance the University's diversity and inclusion goals.

• Consult, advise and collaborate with University officials and offices regarding diversity and inclusion issues and opportunities.

• Enhance recruitment and employment of underrepresented faculty and staff, as well as the recruitment of underrepresented students.

• Assess and report on institutional progress as it relates to diversity and inclusion. Represent the president and Clemson University as needed.

• Provide active leadership to ensure an institutional focus and emphasis on diversity and inclusion.

• Serve as the administrative liaison to campus commissions, committees and boards established to address various dimensions of diversity and inclusion.

The CUCDI oversees more than 30 programs dedicated to improving the climate of diversity, equity and inclusion across the university and between the university and outside communities.

The proposed program will leverage these programs through program governance, design and recruiting. Feedback to the proposed project will be carried out through the Associate Deans for Inclusive Excellence in the College of Engineering, Computing and Applied Science (CECAS) and the College of Behavioral, Social and Health Sciences (CBSHS).

Data Management Plan

Products Of Research

The primary product of this research project will be the MathScy system, a compositional and modular tool integrating advanced AI technologies, including fine-tuned Mixture-of-Expert LLMs and external mathematical tools, to enhance interactive theorem provers (ITPs). This will include the development and release of source code for the MathScy system and associated libraries, enabling conjecture formulation, proof construction, and counterexample finding. Additionally, the project will produce standardized datasets tailored for benchmarking mathematical reasoning systems, formalized representations for mathematical processes, and a suite of significant open problems in mathematics and theoretical computer science for evaluation purposes. Comprehensive documentation and user manuals will be provided to ensure accessibility and usability, along with scholarly publications detailing the research findings and methodologies. These products aim to advance the functionality and applicability of ITPs through Human-Centered and Explainable AI techniques.

Data Format

The data for this project will be formatted to facilitate ease of use, interoperability, and effective benchmarking. Standardized representations will be developed for mathematical problems, conjectures, proofs, and counterexamples, ensuring consistency across datasets. These representations will be structured in machine-readable formats such as JSON and XML, allowing seamless integration with the MathScy system and other mathematical tools. The datasets will include detailed annotations, metadata, and provenance information to provide context and support reproducibility. Additionally, challenge problems and datasets will be encoded in standardized formats compatible with common mathematical software and theorem proving environments. This structured and well-documented data format will ensure that the resources created are accessible, reusable, and valuable for the broader research community.

How Will The Data Be Collected Or Created?

The data for this project will be collected and created through a combination of automated processes, expert input, and collaborative efforts. Initially, existing mathematical datasets and repositories will be mined for relevant problems, conjectures, proofs, and counterexamples. Automated tools will be employed to extract, preprocess, and standardize this information. To supplement these existing resources, new datasets will be generated by leveraging the capabilities of the MathScy system, including conjecture formulation and proof construction features. Expert mathematicians and computer scientists will contribute by curating challenging open problems and providing insights to ensure the quality and relevance of the datasets. Additionally, collaborative efforts with academic institutions and research communities will help gather diverse and significant mathematical challenges. All collected and created data will be meticulously documented, annotated, and stored in standardized, machine-readable formats to facilitate ease of use and reproducibility.

What Documentation And Metadata Will Accompany The Data?

The data will be accompanied by detailed documentation and metadata to ensure usability, transparency, and reproducibility. The documentation will describe the datasets' purposes, scopes, and methodologies, providing guidelines for effective access and use, including examples. Metadata will include dataset descriptions (title, version, date, authors, summary), data provenance (sources and references), data structure (formats, schemas, field descriptions), annotations (explanations of mathematical content), data quality (validation methods, accuracy, biases), usage instructions (access guidelines, software requirements, examples), licensing and access (terms of use, permissions), and contact information (author/maintainer details). This thorough documentation will make the

datasets transparent, understandable, and effectively usable by researchers, developers, and other stakeholders.

How Will You Manage Any Ethical Issues?

To manage ethical issues, this project will prioritize data privacy and security, ensuring compliance with data protection regulations and anonymizing sensitive information. Informed consent will be obtained from human experts involved in data curation, with clear communication about their contributions. Bias mitigation strategies will be implemented, including diverse data sourcing and regular audits to ensure fair representation. Transparency and accountability will be maintained through detailed documentation and regular updates. The project will focus on Human-Centered and Explainable AI, ensuring AI systems are interpretable and understandable. Adherence to ethical standards set by professional bodies will be ensured through regular reviews, and a protocol will be established for addressing any ethical concerns promptly and effectively.

Data Access And Sharing

Data access and sharing policies will promote transparency, collaboration, and ethical use. All datasets, representations, and code will be publicly accessible via an online repository under an open-access license like Creative Commons or MIT, ensuring free use with proper attribution. Access controls will protect sensitive information and comply with privacy regulations for human participant data. Documentation will provide guidelines on accessing, using, and citing resources. Regular updates and communication channels will keep the community informed, and feedback mechanisms will improve the quality and usability of shared resources.

Archiving Of Data

To ensure long-term preservation and accessibility, this project will implement robust data archiving practices. All datasets, code, and documentation will be stored in a secure, centralized repository with version control, such as GitHub or an institutional data repository. Regular backups and storage in multiple locations will ensure data redundancy. The repository will use established archiving standards, including persistent identifiers like DOIs for citation and retrieval. Metadata will be meticulously maintained, and a plan will be developed for periodic review and updates. Access to archived data will be maintained through an open-access policy, benefiting the research community and stakeholders for years to come.

Who Will Be Responsible For Data Management?

The Principal Investigator (PI) and their team will be responsible for data management. The PI will oversee data policies and coordination, while Data Curators will handle the collection, organization, and annotation of datasets. IT Support Staff will manage secure storage, backups, and access control. A Compliance Officer will ensure adherence to legal, ethical, and institutional guidelines, including data privacy regulations. Together, they will ensure effective data management throughout the project lifecycle.

Mentoring Plan To Postdoc/undergraduate Students By The PI

Here's a structured mentoring plan that outlines strategies for fostering the development of both postdoctoral researchers (postdocs) and undergraduate students involved in MathScy.

For Postdoctoral Researchers

Our objective is to develop postdocs into independent researchers and leaders in AI and mathematical theorem proving. This is achieved through the following mentoring activities: (1) Regular One-on-One Meetings: Schedule bi-weekly meetings to discuss progress, address challenges, and set short-term objectives. Provide feedback on research methodology, data analysis, and writing. (2) Professional Development: Encourage attendance at workshops and conferences to present their work and network with other researchers. Provide opportunities to write grant proposals and participate in peer review processes. (3)Leadership and Teaching Opportunities: Assign leadership roles within the project, such as coordinating research teams or managing a part of the project. Offer opportunities to mentor undergraduate students or teach seminars related to their research. (4) Career Counseling: Discuss long-term career goals and options within academia, industry, or governmental agencies. Assist in job application processes, including preparation of CVs, cover letters, and mock interviews. (5) Publication Strategy: Develop a publication plan targeting high-impact journals and conferences. Collaborate on writing and submitting papers, ensuring the postdoc leads multiple publications.

For Undergraduate Students

Our objective is to provide a foundational research experience and enhance skills in AI, formal methods, and mathematical reasoning. This is achieved through (1) Structured Learning Objectives: Define clear, achievable goals for each semester, such as learning specific software tools, completing a literature review, or contributing to a research paper. Provide structured tasks and regular feedback to facilitate learning and development. Use of KANBAN boards. (2) Weekly Group Meetings: Include students in weekly team meetings where they can learn from listening and participating in discussions about the project's progress. Encourage them to ask questions and present their own work periodically. (3)Hands-on Training: Pair each undergraduate with a postdoc or a more experienced graduate student for day-to-day guidance. Organize monthly workshops or tutorials on research methods, ethics in research, data handling, and specific tools used in the MathScy project. (4) Exposure to Professional Environments: Involve them in preparing presentations for team meetings to build confidence and presentation skills. Encourage submission of abstracts to student conferences or symposiums. (5) Feedback and Evaluation: Provide constructive feedback on their performance and progress. Conduct semi-annual reviews to discuss achievements and areas for improvement.

General Considerations

We consider two aspects the most; one is cultural sensitivity and inclusion, where we ensure that mentoring practices are inclusive and sensitive to the diverse backgrounds of all participants. And secondly a general adaptability and agile-ness where we regularly assess the effectiveness of the mentoring plan and make adjustments based on feedback from both postdocs and undergraduates.